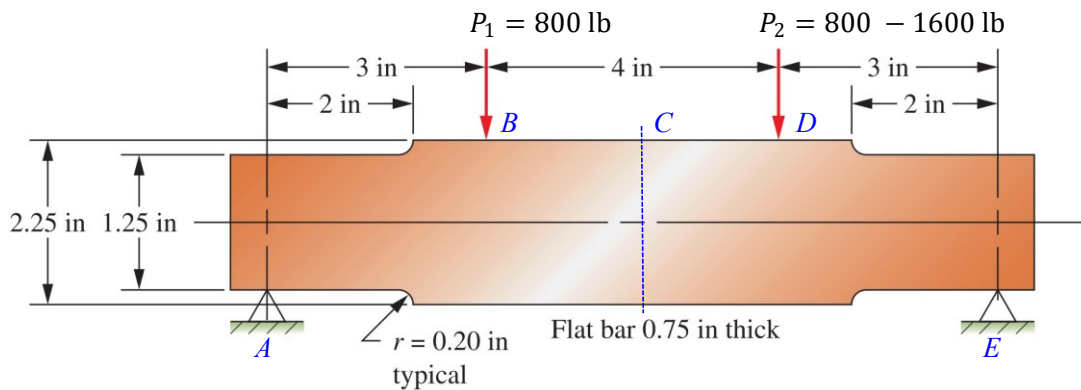


Machine Design (EMCH327 Section 001, Spring 2022)

Mid-Term Exam 2, 8:30 AM, Wednesday, Mar. 16th, 2022

Name (First/Last): _____ Score: _____/100'

- For a beam shown below, the left load $P_1 = 800$ lb remains constantly applied, while the right load P_2 is applied and changes from 800 lb to 1600 lb as the machine cycles. Determine the stress ratio **for the point D**. Also, for the same point, use the maximum shear stress theory to determine the design factor so that the aluminium alloy 5052-H34 can be safely selected as material for the beam. (40' in total)

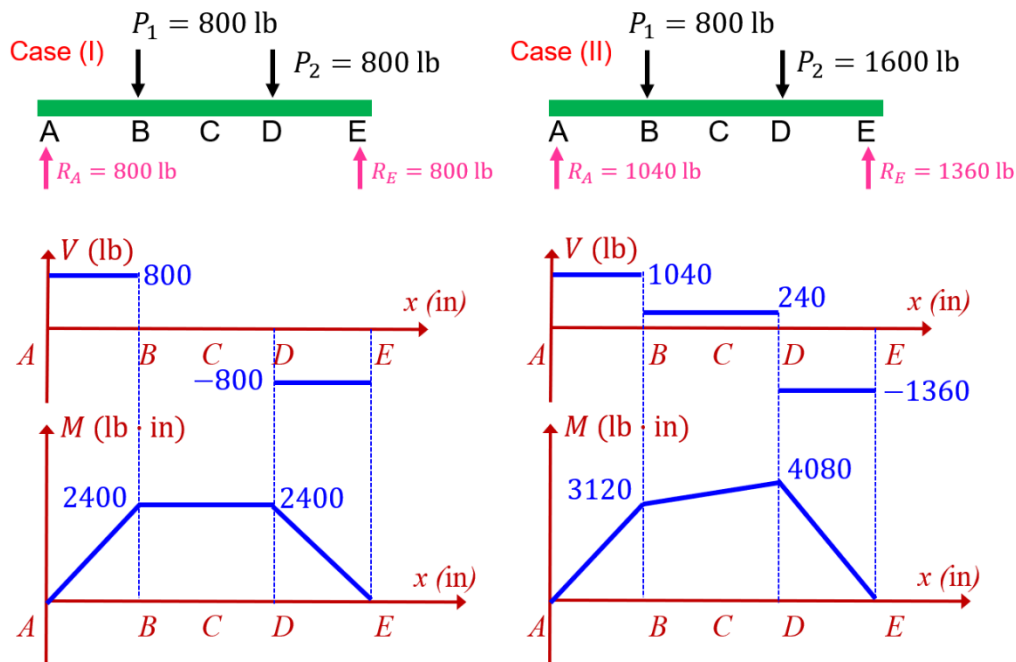


Solutions:

Step 1: Determine the maximum and minimum stress for the middle point D.

There are two loading cases: (I) $P_2 = 800$ lb; (II) $P_2 = 1600$ lb, both with $P_1 = 800$ lb.

Using Free-Body-Diagrams to solve the reactions at two ends and force and moment diagrams:



For the point D , the maximum bending moment for the loading case (I) and (II) are $2400 \text{ lb} \cdot \text{in}$ and $4080 \text{ lb} \cdot \text{in}$, respectively.

With rectangular cross section, according to Appendix 1, the bending modulus

$$S = \frac{BH^2}{6} = \frac{0.75(2.25)^2}{6} = 0.6328 \text{ in}^3$$

The maximum and minimum stresses during the cyclic loading are

$$\sigma_{\max} = \frac{M_{\max}}{S} = \frac{4080}{0.6328} = 6448 \text{ psi}$$

$$\sigma_{\min} = \frac{M_{\min}}{S} = \frac{2400}{0.6328} = 3793 \text{ psi}$$

$$\text{Stress ratio } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{3793}{6448} = \boxed{0.588}$$

Step 2: The case (II) induces the highest possible normal tensile stress on the surface of the beam by the maximum bending moment ($4080 \text{ lb} \cdot \text{in}$), which is also 2D plane stress state. The corresponding square stress state and Mohr's circle can be drawn.

With $\sigma_x = \sigma_{\max} = 6448 \text{ psi}$, $\sigma_y = \tau_{xy} = 0$, we know the two principal stresses

$$\sigma_1 = \sigma_x = 6448 \text{ psi}, \sigma_2 = \sigma_3 = 0,$$

and the absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \frac{6448}{2} = 3224 \text{ psi}$$

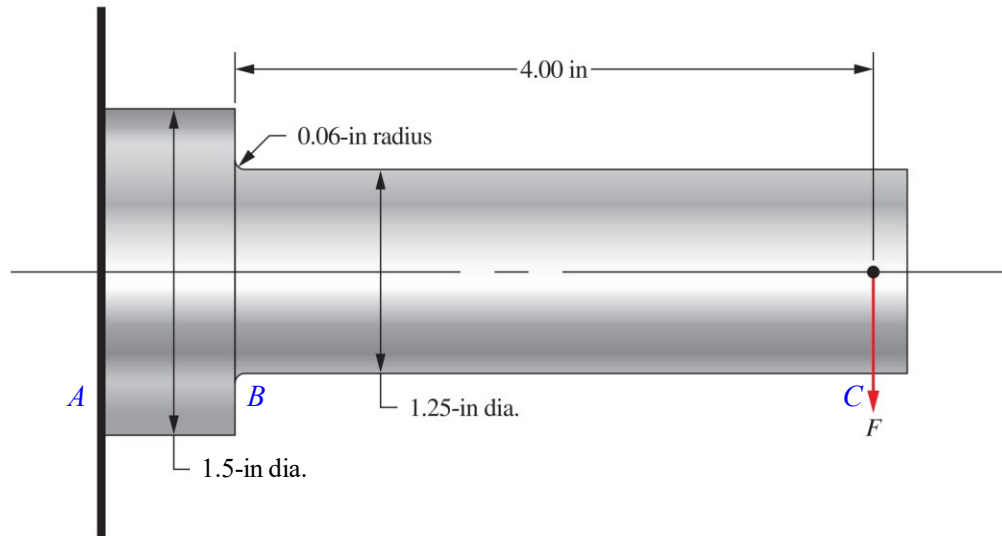
The Maximum Shear Stress Theory (MSST) says that a material with tensile yield strength should fulfill

$$\frac{s_y}{N} \geq \sigma_1 - \sigma_3$$

For alloy 5052-H34, from Appendix 9, we find that $s_y = 31 \text{ ksi}$. Then,

$$N \leq \frac{s_y}{\sigma_1} = \frac{31000}{6448} = \boxed{4.8}$$

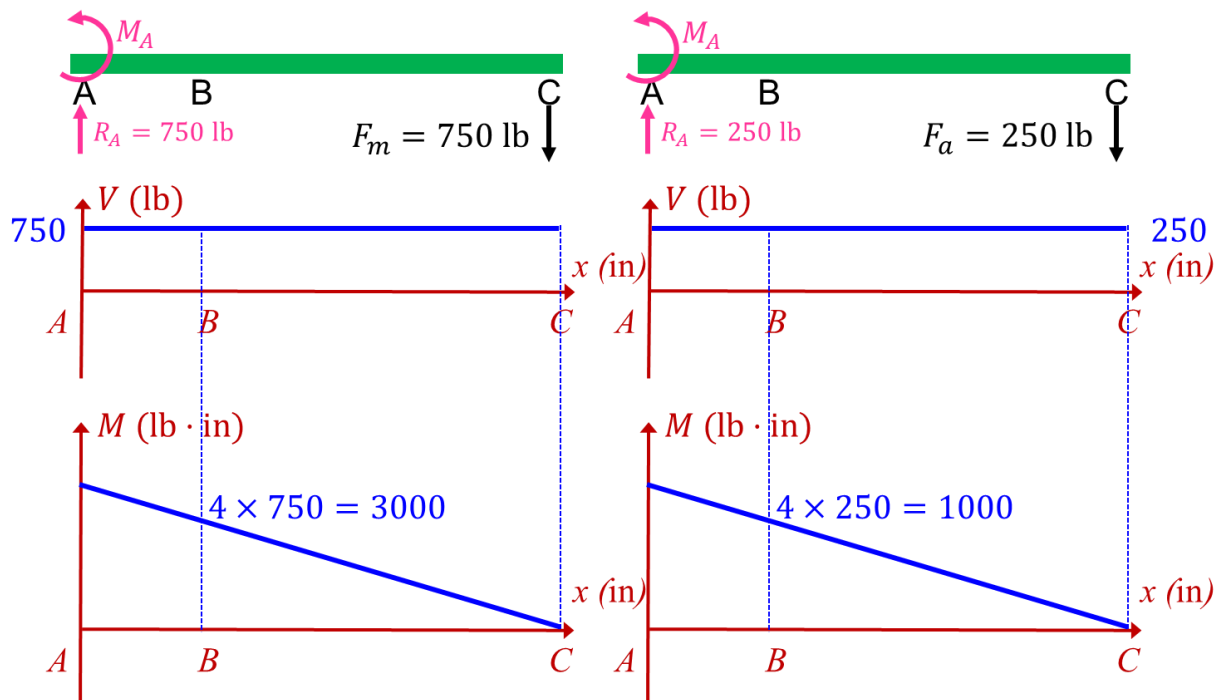
2. Use Goodman criterion to determine the design factor for a shaft with circular cross-section to carry a downward load that varies from 500 to 1000 lb. The material is wrought steel with $s_u = 96$ ksi and $s_n = 48$ ksi. Take reliability as 90%. Stress concentration needs to be considered at point B. (30' in total)



Solutions:

Step 1: Since the loading varies, meaning that we need to design the shaft under cyclic loadings. To this end, we first need to determine the maximum and minimum stress in the shaft over time.

Using Free-Body-Diagrams to solve the reactions at end A and then plot the force and moment diagrams.



Note that, the downward load F varies from 500 to 1000 lb. Thus, the mean and alternating loading (i.e., the amplitude of loading) is

$$F_m = \frac{500 + 1000}{2} = 750 \text{ lb}$$

$$F_a = \frac{1000 - 500}{2} = 250 \text{ lb}$$

The bending moment occurs at the point B of the shaft

$$M_B = 3000 \text{ lb} \cdot \text{in for } F_m = 750 \text{ lb}$$

$$M_B = 1000 \text{ lb} \cdot \text{in for } F_a = 250 \text{ lb}$$

Then, the corresponding mean and alternating stress is

$$\sigma_m = \frac{M_B|_{F_m}}{S} = \frac{3000}{\pi \times 1.25^3 / 32} = 15646 \text{ psi}$$

$$\sigma_a = \frac{M_B|_{F_a}}{S} = \frac{1000}{\pi \times 1.25^3 / 32} = 5215 \text{ psi}$$

Step 2: To use Goodman criterion, i.e., $\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t \sigma_a}{s_n'}$, we need to know the estimated actual endurance limit (s_n'). Use Eq. (5-21) in the textbook to determine s_n' :

$$s_n' = s_n \cdot C_m \cdot C_{st} \cdot C_R \cdot C_S = 48(1.0)(1.0)(0.90)(0.95) = 41.04 \text{ ksi}$$

Wrought steel: $C_m = 1.0$

Bending stress: $C_{st} = 1.0$

Reliability factor: $C_R = 0.90$ (Table 5-3)

Size factor: $C_S = 0.95$ (Table 5-4 using equivalent diameter 0.4625 in)

(For nonrotating circular section in repeated and reversed flexure, the equivalent diameter is $D_e = 0.37D = 0.4625 \text{ in}$)

Step 3: At fillets (point B) determine stress concentration factor:

$$r/d = 0.06/1.25 = 0.048 \approx 0.05$$

$$D/d = 1.5/1.25 = 1.2$$

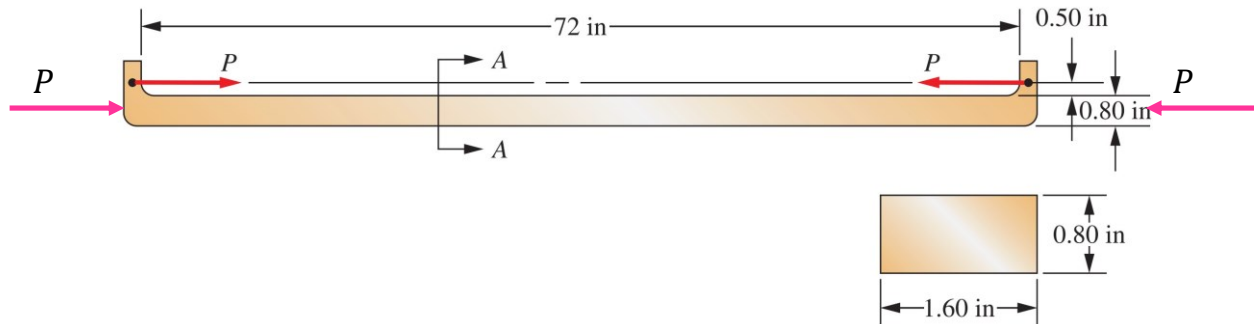
From Appendix Figure A18-6, we get the stress concentration factor $K_t = 2.05$

Finally, using Goodman criterion,

$$\frac{1}{N} = \frac{15646}{96000} + \frac{2.05 \times 5215}{41040} = 0.4235$$

$N = 2.4$

3. The column shown is subjected to opposing forces. The material is steel SAE 1117 WQT 350. First, determine the maximum allowable **centrical** load to achieve a design factor of 3. Second, determine whether the column is still safe when the same allowable load is **eccentrically** applied. (30' in total)



Solutions:

For centrically loaded column:

Step 1:

From Appendix 1, the minimum radius of gyration is $r = \sqrt{I/A} = \frac{B}{\sqrt{12}} = \frac{0.8}{\sqrt{12}} = 0.231$

Step 2:

Slenderness Ratio $S_r = \frac{KL}{r} = \frac{1.0 \times 72}{0.231} = 311.7$

Step 3:

The material is steel SAE 1117 WQT 350. From Appendix 3, we find the yield strength $s_y = 50$ ksi and stiffness (Young's modulus) $E = 30$ Mpsi.

Step 4:

The Euler-Johnson transition slenderness ratio $C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2 (30 \times 10^6)}{50 \times 10^3}} = 108.8$.

Since $S_r > C_c$, the column is long column (Euler column).

Step 5:

Using Euler formula Eq. (6-4), we get the critical central load

$$P_{cr} = \frac{\pi^2 EA}{S_r^2} = \frac{\pi^2 \times 30 \times 10^6 \times 0.8 \times 1.6}{311.7^2} = 3901 \text{ lb}$$

Step 6:

The corresponding allowable load $P_a = \frac{P_{cr}}{N} = \frac{3901}{3} = 1300 \text{ lb}$

For eccentrically loaded column:

Cross-section area $A = B \cdot H = (0.8)(1.6) = 1.28 \text{ in}^2$

Radius of gyration is $r = \sqrt{I/A} = \frac{B}{\sqrt{12}} = \frac{0.8}{\sqrt{12}} = 0.231$

Eccentricity $e = 0.5 + \frac{0.8}{2} = 0.9 \text{ in}$

Distance from neutral axis to outside of section $c = \frac{B}{2} = \frac{0.8}{2} = 0.4 \text{ in}$

Then, using Eq. (6-15) we get the required yield strength

$$\begin{aligned} s_y|_{\text{req.}} &= \frac{NP_a}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{KL}{2r} \sqrt{\frac{NP_a}{AE}} \right) \right] \\ &= \frac{3(1300)}{1.28} \left[1 + \frac{(0.9)(0.4)}{0.231^2} \sec \left(\frac{1.0(72)}{2(0.231)} \sqrt{\frac{3(1300)}{1.28(30 \times 10^6)}} \right) \right] \\ &= 23610 \text{ psi} \end{aligned}$$

Since $s_y|_{\text{req.}} < s_y = 50 \text{ ksi} = 50000 \text{ psi}$, the column **is still safe** for the same eccentric loading with off-distance 0.9 in.

*****Bonus Problem***** Using the boundary-lubricated approach, for a plain surface bearing to carry a radial load of 100 lb from a shaft having a diameter of 0.80 in and rotating at 800 rpm, determine the **design value** of pV factor. Use an L/D ratio of the bearing to be 1.25. (**10'** in total)

Solutions:

Given $D = 0.80$ in and $L/D = 1.25$, then the nominal length of the bearing is

$$L = 1.25D = 1.25(0.80) = 1.00 \text{ in}$$

The bearing pressure (using the projected area of the bearing) is

$$p = \frac{F}{LD} = \frac{100}{(1.00)(0.8)} = 125 \text{ psi}$$

The linear speed of the journal surface is

$$V = \frac{\pi Dn}{12} = \frac{\pi(0.8)(800)}{12} = 167.55 \text{ ft/min}$$

The pV factor is

$$pV = (125)(167.55) = 20945 \text{ psi} \cdot \text{fpm}$$

$$\text{Design value of } pV = 2(20945) = 41890 \text{ psi} \cdot \text{fpm}$$